

When Healthy-Only Transfer Fails in PMSM Stator-Fault Detection: A Leakage-Resistant Cross-Dataset Evaluation

Online Resource 1 for Journal of Electrical Engineering & Technology - anonymous review copy

Submission draft generated mechanically from hash-locked CSV/JSON artifacts. The frozen external experiment uses one physical motor. Sections S3-S7 are explicitly post-reveal diagnostics and did not select a new model or threshold. Section S9 preserves a prospectively logged parser failure and labels the repaired 200 W analysis as post-reveal sensitivity.

Technical summary

The complete frozen external comparison favors Target MinCovDet, not Proposed: Target MinCovDet detected 70.05% of 384 fault blocks with 0/32 held-out healthy false alarms, whereas Proposed detected 25.00% with 1/32 false alarms. Conditional complete-record comparisons show that adding source healthy data generally reduced external performance, except that target-only and source+target OC-SVM tied in mean detection. The apparent performance increase late in each record is coupled to the shared acceleration ramp. Five-seed, feature-geometry, and 100 kHz-to-10 kHz sensitivity analyses did not change the frozen primary result. All external intervals and post-reveal comparisons remain conditional on a single physical motor. The secondary 21-record transient audit did not supply confirmatory evidence: the frozen parser accepted 0/21 records, the repaired 20 kW arm failed its compatibility gate, and all methods had zero first-second detection in the repaired 200 W sensitivity.

S1. Data, experimental units, and locked splits

The **physical motor** is the generalization unit, the **MAT/TDMS record** is the condition-level resampling unit, the 3 s **block** is the alarm unit, and the 0.2 s window is the feature-extraction unit. Repeated windows and ordered blocks from one record are not independent replications. The external dataset contains only one physical motor; its 48 fault records are operating-condition records, not 48 motors.

Dataset	Physical motors	Physical records	Record-subsystem streams	Sampling	Frozen analysis grain
KAIST source	3 (1, 1.5, and 3 kW)	45 (3 healthy; 42 fault)	not applicable	100 kHz; leading 120 s	600 non-overlapping 0.2 s windows and 40 3 s blocks per record
External dual-three-phase PMSM	1 (custom 30.16 kW)	56 (8 healthy; 48 fault)	112 (2 subsystems per record)	10 kHz; [12, 36) s	120 windows and 8 blocks per record-subsystem; one system block after maxima

The external MAT records contain two three-phase current subsystems (SubSys1 and SubSys2). Each subsystem is scored separately. A block score is the maximum over 15 windows within a subsystem followed by the maximum over the two subsystems. Thus, each physical record contributes exactly eight system block scores.

S1.1. Ordered split definitions

Stage	KAIST leave-one-motor-out	External validation
Source reference	healthy blocks 0-23 for each of the two source motors; balanced subset 0, 8, 15, 23	three KAIST source motors; covariance blocks 0-23 and balanced blocks 0, 8, 15, 23
Target adaptation	blocks 0-3 (4 blocks; 60 windows) from the held-out motor's healthy record	0 N m healthy record, blocks 0-3 (60 windows per subsystem)
Guard / unused	blocks 4 and 25	0 N m blocks 4-7 are unused for validation
Calibration	healthy blocks 5-24 (20 blocks)	healthy loads 10, 20, and 30 N m; all 8 blocks (3 records; 24 system blocks)
Held-out health test	healthy blocks 26-39 (14 blocks)	healthy loads 5, 15, 25, and 35 N m; all 8 blocks (4 records; 32 system blocks)
Fault test	14 records $\times$ 40 blocks = 560 blocks per target motor	6 turn-fault states $\times$ 8 loads = 48 records; 8 blocks each = 384 system blocks

The external 6-by-8 grid is not crossed by fault phase. Fault-turn levels 1, 3, 5, and 6 use phase U, whereas levels 2 and 4 use phase V. Consequently, fault-turn count and phase effects are not separately identifiable; the turn-stratified record bootstrap conditions on this fixed assignment rather than resolving it.

All models use the 27-feature scale-free arm: fundamental_hz, sequence_unbalance, fundamental_amplitude_cv, phase_rms_cv, clarke_radius_cv, zero_sequence_ratio, spectral_entropy, sideband_lower_ratio, sideband_upper_ratio, harmonic_2_ratio_mean, harmonic_2_ratio_max, harmonic_3_ratio_mean, harmonic_3_ratio_max, harmonic_4_ratio_mean, harmonic_4_ratio_max, harmonic_5_ratio_mean, harmonic_5_ratio_max, thd_2_to_5_mean, rms_ratio_a, crest_a, kurtosis_a, rms_ratio_b, crest_b, kurtosis_b, rms_ratio_c, crest_c, kurtosis_c. Source features are robustly centered and scaled within source motor; external target features are robustly centered and scaled within subsystem from the frozen adaptation subset. The split-conformal alarm level is $\alpha = 0.05$. Arithmetic covariance methods use ridge fraction 0.001 and the Proposed Log-Euclidean method uses the source-selected ridge fraction 0.01. No external fault row enters fitting, scaling, calibration, or threshold construction.

S2. Complete frozen external comparison across 11 methods

Target MinCovDet was the strongest external method: 70.05% fault-block detection, zero false alarms among 32 held-out healthy blocks, and AUROC 0.9268. It was also the only method satisfying the prespecified empirical H1 rule. Proposed achieved 25.00% detection, one false alarm (3.12%), and AUROC 0.6354. These external results are reported in full below; method-specific score thresholds are not comparable in magnitude across estimators.

The external pooled AUROC compares 384 fault blocks spanning all eight loads with 32 held-out health blocks from only 5, 15, 25, and 35 N m. It is therefore a descriptive ranking statistic under unequal load support, not a load-matched population estimand.

Method	Healthy training/reference	Threshold	Health FA	H1	Fault block detection	Record-macro detection	Record-bootstrap 95% CI (%)	Record any-alarm	AUROC	AUPRC
Target Ledoit-Wolf	target adaptation	183.887	1/32 (3.12%)	fail	141/384 (36.72%)	36.72%	[32.55, 41.41]	100.00%	0.7675	0.9747
Target sample covariance	target adaptation	738.703	1/32 (3.12%)	fail	215/384 (55.99%)	55.99%	[51.04, 61.46]	100.00%	0.8757	0.9883
Source covariance	source healthy	109212	1/32 (3.12%)	fail	49/384 (12.76%)	12.76%	[12.50, 13.28]	100.00%	0.5747	0.9435
Entity-balanced covariance	source entities + target	1250.37	1/32 (3.12%)	fail	105/384 (27.34%)	27.34%	[23.18, 32.03]	100.00%	0.6972	0.9648
Proposed: Log-Euclidean entity covariance	source entities + target	4983.29	1/32 (3.12%)	fail	96/384 (25.00%)	25.00%	[21.09, 29.43]	100.00%	0.6354	0.9583
Target OC-SVM (RBF)	target adaptation	0.401736	1/32 (3.12%)	fail	49/384 (12.76%)	12.76%	[10.68, 15.36]	87.50%	0.7209	0.9622
Source+target OC-SVM (RBF)	source-balanced + target	0.827663	1/32 (3.12%)	fail	49/384 (12.76%)	12.76%	[11.20, 14.58]	91.67%	0.6876	0.9558
Target Isolation Forest	target adaptation	0.569218	3/32 (9.38%)	fail	230/384 (59.90%)	59.90%	[57.29, 62.76]	100.00%	0.8670	0.9876
Source+target Isolation Forest	source-balanced + target	0.581677	2/32 (6.25%)	fail	194/384 (50.52%)	50.52%	[47.92, 53.12]	100.00%	0.8245	0.9833
Target MinCovDet	target adaptation	1238.1	0/32 (0.00%)	pass	269/384 (70.05%)	70.05%	[65.36, 75.26]	100.00%	0.9268	0.9934
Source+target MinCovDet	source-balanced + target	9802.52	1/32 (3.12%)	fail	116/384 (30.21%)	30.21%	[26.04, 34.90]	100.00%	0.7013	0.9666

H1 requires both a descriptive Wilson upper bound no greater than 0.12 and a maximum held-out-load false-alarm rate no greater than 0.15. The record-bootstrap interval resamples complete fault records within the six fault-turn strata (10,000 draws, seed 711). Because every fault record has eight blocks, block-weighted and record-macro detection coincide here. Intervals are conditional on the observed single external motor.

S3. Conditional paired comparisons and source-transfer contrasts

S3.1. Every frozen candidate versus Proposed

Candidate	Candidate detection	Proposed detection	Candidate - Proposed	Paired 95% CI (pp)	Bootstrap p	Holm p	Candidate / tie / Proposed records	Candidate / Proposed health FA
Target MinCovDet	70.05%	25.00%	+45.05 pp	[+40.62, +49.48]	0.0001	0.0010	43 / 5 / 0	0 / 1
Target Isolation Forest	59.90%	25.00%	+34.90 pp	[+31.77, +38.02]	0.0001	0.0010	47 / 1 / 0	3 / 1
Target sample covariance	55.99%	25.00%	+30.99 pp	[+28.12, +34.38]	0.0001	0.0010	48 / 0 / 0	1 / 1
Source+target Isolation Forest	50.52%	25.00%	+25.52 pp	[+21.88, +28.91]	0.0001	0.0010	41 / 7 / 0	2 / 1
Target Ledoit-Wolf	36.72%	25.00%	+11.72 pp	[+9.90, +13.54]	0.0001	0.0010	33 / 15 / 0	1 / 1
Source+target MinCovDet	30.21%	25.00%	+5.21 pp	[+3.91, +6.51]	0.0001	0.0010	21 / 26 / 1	1 / 1
Entity-balanced covariance	27.34%	25.00%	+2.34 pp	[+1.04, +3.65]	0.0001	0.0010	10 / 36 / 2	1 / 1
Source+target OC-SVM (RBF)	12.76%	25.00%	-12.24 pp	[-16.15, -8.59]	0.0001	0.0010	0 / 23 / 25	1 / 1
Source covariance	12.76%	25.00%	-12.24 pp	[-16.41, -8.33]	0.0001	0.0010	0 / 23 / 25	1 / 1
Target OC-SVM (RBF)	12.76%	25.00%	-12.24 pp	[-16.41, -8.59]	0.0001	0.0010	0 / 22 / 26	1 / 1

S3.2. Target-only versus source-balanced training/reference

Contrast family	Target-only	Source-balanced counterpart	Target detection	Balanced detection	Target - balanced	Paired 95% CI (pp)	Holm p	Target / tie / balanced records
min cov det	Target MinCovDet	Source+target MinCovDet	70.05%	30.21%	+39.84 pp	[+35.42, +44.27]	0.0005	42 / 6 / 0
sample vs log euclidean entity covariance	Target sample covariance	Proposed: Log-Euclidean entity covariance	55.99%	25.00%	+30.99 pp	[+28.12, +34.38]	0.0005	48 / 0 / 0
sample vs arithmetic entity covariance	Target sample covariance	Entity-balanced covariance	55.99%	27.34%	+28.65 pp	[+25.78, +32.03]	0.0005	47 / 1 / 0
isolation forest	Target Isolation Forest	Source+target Isolation Forest	59.90%	50.52%	+9.38 pp	[+7.29, +11.46]	0.0005	32 / 15 / 1
ocsvm	Target OC-SVM (RBF)	Source+target OC-SVM (RBF)	12.76%	12.76%	+0.00 pp	[-1.04, +1.04]	1.0000	2 / 44 / 2

Both tables use complete-record paired differences across the same 48 fault records, stratified by fault turns. Two-sided centered bootstrap p-values use 10,000 draws (seed 711); Holm adjustment is applied separately to the 10 candidate-vs-Proposed comparisons and the five transfer contrasts. The smallest attainable nonzero bootstrap estimate is 0.0001. These post-reveal comparisons quantify conditional differences on one motor; their adjusted p-values do not establish cross-motor generalization. The two covariance contrasts against target sample covariance are the closest transfer comparisons but do not hold the matrix estimator fixed.

S4. Horizon, first-alarm, and acceleration-position diagnostics

All records follow the same [12, 36] s acceleration segment, divided into block IDs 0-7. Block position therefore co-moves with speed. The primary all-eight-block metric gives late high-speed blocks more opportunities to alarm; the following post-reveal horizon cuts retain the frozen scores and thresholds.

S4.1. First four, first seven, and all eight blocks

Method	Blocks 0-3: detection / any / health FA (%)	Blocks 0-6: detection / any / health FA (%)	Blocks 0-7: detection / any / health FA (%)
Target Ledoit-Wolf	6.77 / 16.67 / 0.00	27.68 / 87.50 / 0.00	36.72 / 100.00 / 3.12
Target sample covariance	25.00 / 35.42 / 0.00	49.70 / 100.00 / 0.00	55.99 / 100.00 / 3.12
Source covariance	0.00 / 0.00 / 0.00	0.30 / 2.08 / 0.00	12.76 / 100.00 / 3.12
Entity-balanced covariance	3.12 / 10.42 / 0.00	16.96 / 62.50 / 0.00	27.34 / 100.00 / 3.12
Proposed: Log-Euclidean entity covariance	2.60 / 10.42 / 0.00	14.29 / 52.08 / 0.00	25.00 / 100.00 / 3.12
Target OC-SVM (RBF)	0.00 / 0.00 / 0.00	2.08 / 6.25 / 0.00	12.76 / 87.50 / 3.12
Source+target OC-SVM (RBF)	0.00 / 0.00 / 0.00	1.49 / 6.25 / 0.00	12.76 / 91.67 / 3.12
Target Isolation Forest	29.17 / 85.42 / 18.75	54.46 / 100.00 / 10.71	59.90 / 100.00 / 9.38
Source+target Isolation Forest	19.79 / 75.00 / 12.50	44.35 / 97.92 / 7.14	50.52 / 100.00 / 6.25
Target MinCovDet	54.69 / 64.58 / 0.00	65.77 / 89.58 / 0.00	70.05 / 100.00 / 0.00
Source+target MinCovDet	5.21 / 14.58 / 0.00	20.24 / 62.50 / 0.00	30.21 / 100.00 / 3.12

Each cell reports fault-block detection, fault-record any-alarm, and held-out healthy block false-alarm rate. Denominators are respectively 192/48/16 for blocks 0-3, 336/48/28 for blocks 0-6, and 384/48/32 for blocks 0-7.

S4.2. First alarm by physical fault record

Method	Records ever alarmed	Never alarmed	First block Q1 / median / Q3	Healthy-RPM proxy Q1 / median / Q3
Target Ledoit-Wolf	48/48	0	4 / 5 / 6	889 / 1216 / 1643
Target sample covariance	48/48	0	2 / 4 / 5	463 / 889 / 1216
Source covariance	48/48	0	7 / 7 / 7	2214 / 2214 / 2214
Entity-balanced covariance	48/48	0	5 / 6 / 7	1216 / 1643 / 2214
Proposed: Log-Euclidean entity covariance	48/48	0	6 / 6 / 7	1642 / 1643 / 2214
Target OC-SVM (RBF)	42/48	6	7 / 7 / 7	2214 / 2214 / 2214
Source+target OC-SVM (RBF)	44/48	4	7 / 7 / 7	2214 / 2214 / 2214
Target Isolation Forest	48/48	0	0 / 2 / 3	218 / 463 / 644
Source+target Isolation Forest	48/48	0	0 / 1 / 3.2	218 / 340 / 706
Target MinCovDet	48/48	0	0 / 1 / 5	218 / 318 / 1216
Source+target MinCovDet	48/48	0	5 / 6 / 7	1216 / 1643 / 2214

First-alarm quantiles are calculated only among records that ever alarm. The RPM quantity is the median RPM from the matching-load healthy record and block; it is a proxy, not a measured fault-record RPM and not a stationary time-to-detection estimate. Proposed first alarm occurred at median block 6, while Target MinCovDet first alarm occurred at median block 1.

S4.3. AUROC matched by acceleration block

Method	B0	B1	B2	B3	B4	B5	B6	B7	Equal-weight mean	Pooled
Target Ledoit-Wolf	0.385	0.922	1.000	1.000	1.000	1.000	1.000	1.000	0.913	0.767
Target sample covariance	0.844	0.906	0.958	0.990	1.000	1.000	1.000	1.000	0.962	0.876
Source covariance	0.703	0.896	0.948	0.995	1.000	1.000	1.000	1.000	0.943	0.575
Entity-balanced covariance	0.589	0.917	1.000	1.000	1.000	1.000	1.000	1.000	0.938	0.697
Proposed: Log-Euclidean entity covariance	0.526	0.453	0.734	0.854	0.911	0.958	1.000	1.000	0.805	0.635
Target OC-SVM (RBF)	0.464	0.823	0.953	0.990	1.000	1.000	1.000	0.891	0.890	0.721
Source+target OC-SVM (RBF)	0.464	0.802	0.948	0.990	0.995	1.000	1.000	0.917	0.889	0.688
Target Isolation Forest	0.453	0.724	0.979	0.964	0.974	0.984	1.000	1.000	0.885	0.867
Source+target Isolation Forest	0.505	0.807	0.885	0.990	0.990	1.000	1.000	1.000	0.897	0.825
Target MinCovDet	0.755	0.880	0.969	1.000	1.000	1.000	1.000	1.000	0.951	0.927
Source+target MinCovDet	0.698	0.917	1.000	0.969	0.932	0.979	1.000	1.000	0.937	0.701

Blocks B0-B7 correspond to [12,15), [15,18), [18,21), [21,24), [24,27), [27,30), [30,33), and [33,36) s. Each position-specific AUROC compares four held-out healthy blocks with 48 fault blocks at the same position. The equal-weight mean describes matched-position discrimination; the pooled AUROC remains the frozen primary ranking metric. Neither treats the eight ordered blocks as independent motors.

S5. Five-seed stochastic sensitivity

Stochastic method	Primary detection	Five-seed detection	Range	Max change from primary	Health FA range	H1 passes	Threshold range	AUROC range
Target Isolation Forest	59.90%	51.30-59.90%	8.59 pp	8.59 pp	1-3/32 (3.12-9.38%)	0/5	0.562858-0.569218	0.8371-0.8906
Source+target Isolation Forest	50.52%	47.92-53.65%	5.73 pp	3.12 pp	0-3/32 (0.00-9.38%)	2/5	0.578661-0.593532	0.8062-0.8530
Target MinCovDet	70.05%	65.36-77.08%	11.72 pp	7.03 pp	0-2/32 (0.00-6.25%)	3/5	934.173-1396.26	0.9219-0.9352
Source+target MinCovDet	30.21%	30.21-33.85%	3.65 pp	3.65 pp	1-1/32 (3.12-3.12%)	0/5	8857.36-9802.52	0.6795-0.7320

The frozen seeds are 20260820, 1201, 2402, 3603, 4804; 20260820 is the primary seed. For each seed, the estimator is refitted only on the same frozen healthy training subset, and its threshold is mechanically recalibrated from the same 24 healthy calibration blocks at alpha 0.05. No fault score affects a threshold. Primary-seed reconciliation gave zero alarm mismatches for all four methods (maximum score discrepancy 1.46×10^{-11}). No formal seed-stability gate was specified before reveal, so the full ranges are descriptive and were not used to change the reported model or protocol.

S6. Post-reveal feature drift and frozen-geometry reconstruction

The feature diagnostic is descriptive and uses record-subsystem-block means (15 windows). The table shows the eight largest direction-free single-feature AUROC's on held-out health loads, comparing 384 fault blocks with 64 record-subsystem healthy blocks at matching loads and block positions. One healthy condition is reused across six fault-turn comparisons, so the 384 pairs are not independent motor replications.

Feature	Family	Raw direction-free AUROC	Robust-z direction-free AUROC	Matched robust-z dz	Fault contribution share	Health contribution share	Max speed rho
harmonic_3_ratio_max	harmonic sideband	0.927	0.933	+1.032	0.35%	0.71%	0.509
harmonic_3_ratio_mean	harmonic sideband	0.925	0.925	+0.992	0.24%	0.52%	0.501
rms_ratio_a	current shape	0.815	0.862	+0.679	21.62%	3.86%	0.469
phase_rms_cv	sequence imbalance	0.769	0.825	+0.662	9.42%	8.08%	0.513
fundamental_amplitude_cv	current shape	0.769	0.797	+0.661	3.34%	1.26%	0.598
thd_2_to_5_mean	harmonic sideband	0.788	0.795	+0.840	1.21%	0.96%	0.588
rms_ratio_c	current shape	0.664	0.741	-0.179	2.30%	0.45%	0.080
zero_sequence_ratio	sequence imbalance	0.695	0.701	-0.998	8.50%	8.79%	0.067

Harmonic-3 ratios discriminate fault from health individually but receive less than 1% of the frozen score's absolute contribution, whereas `rms_ratio_a` receives 21.62%. This mismatch is a diagnostic of the locked multivariate geometry, not a license to reweight features after reveal. Block and fundamental frequency co-move, and the reported speed associations do not identify causal fault features.

S6.1. Exact reconstruction checks

Subsystem	Adaptation windows	Rank / 27	Condition number	Max contribution-sum error	Center match	Scale match
SubSys1	60	27	309.53	2.910e-11	yes	yes
SubSys2	60	27	300.11	7.276e-11	yes	yes

Across all 448 physical-record block scores, the maximum absolute reconstructed- versus-frozen score error was 3.638e-12; the maximum feature-contribution sum error was 7.276e-11. The reconstructed center, scale, Log-Euclidean covariance, winning subsystem/window, system maximum, p-value, and alarm were checked without fitting a new model or recomputing a threshold. Fault labels did not enter covariance reconstruction.

S7. Post-reveal 100 kHz to 10 kHz sampling-rate sensitivity

The KAIST three-phase current was mechanically anti-aliased and decimated from 100 kHz to 10 kHz with `scipy.signal.resample_poly` (up=1, down=10, Kaiser beta 5.0, line padding) over each complete leading 120 s record before the frozen 0.2 s windows were cut. The feature definitions, 3 s maximum block aggregation, splits, ridge values, methods, seeds, calibration rule, and external feature table were unchanged. This analysis was specified after fault reveal and was not used for model or protocol selection.

Source feature arm	Rows	Records	Windows / record	Blocks / record	Duplicate keys	Nonfinite values
native_100khz	27000	45	600-600	40-40	0	0
polyphase_10khz	27000	45	600-600	40-40	0	0

Method	Sampling dependency	Detection, 100 kHz source	Detection, 10 kHz source	Change	Health FAR, 100 kHz	Health FAR, 10 kHz	AUROC, 100 kHz	AUROC, 10 kHz
Target Ledoit-Wolf	target-only control	36.72%	36.72%	+0.00 pp	3.12%	3.12%	0.7675	0.7675
Target sample covariance	target-only control	55.99%	55.99%	+0.00 pp	3.12%	3.12%	0.8757	0.8757
Source covariance	source-sensitive	12.76%	12.50%	-0.26 pp	3.12%	3.12%	0.5747	0.5496
Entity-balanced covariance	source-sensitive	27.34%	25.26%	-2.08 pp	3.12%	6.25%	0.6972	0.6849
Proposed: Log-Euclidean entity covariance	source-sensitive	25.00%	24.74%	-0.26 pp	3.12%	3.12%	0.6354	0.6331
Target OC-SVM (RBF)	target-only control	12.76%	12.76%	+0.00 pp	3.12%	3.12%	0.7209	0.7209
Source+target OC-SVM (RBF)	source-sensitive	12.76%	12.50%	-0.26 pp	3.12%	3.12%	0.6876	0.6977
Target Isolation Forest	target-only control	59.90%	59.90%	+0.00 pp	9.38%	9.38%	0.8670	0.8670
Source+target Isolation Forest	source-sensitive	50.52%	52.08%	+1.56 pp	6.25%	3.12%	0.8245	0.8643
Target MinCovDet	target-only control	70.05%	70.05%	+0.00 pp	0.00%	0.00%	0.9268	0.9268

Method	Sampling dependency	Detection, 100 kHz source	Detection, 10 kHz source	Change	Health FAR, 100 kHz	Health FAR, 10 kHz	AUROC, 100 kHz	AUROC, 10 kHz
Source+target MinCovDet	source-sensitive	30.21%	27.60%	-2.60 pp	3.12%	3.12%	0.7013	0.6966

Proposed detection changed from 25.00% to 24.74% (-0.26 percentage points), and AUROC changed from 0.6354 to 0.6331. Across the 48 physical fault records, alarm rate improved for 0, tied for 47, and worsened for 1. Its KAIST leave-one-motor-out source detection remained 91.96% (100 kHz: 95.71%) and AUROC remained 0.9947 (100 kHz: 0.9993), so the 10 kHz arm was not globally unusable. All five target-only controls reproduced 448/448 block scores, p-values, and alarms with zero mismatches. These results do not support source/target sampling-rate mismatch as the primary explanation for the observed negative transfer.

S8. Hash registry, interpretation limits, and next validation step

Role	Artifact	SHA-256
processed input	data/processed/kaist_current_features.csv.gz	6276bfae67169f0e70a3b1bd3c4fdbf42c08631a4d4dcfd097ef8246ef970d20
processed input	data/processed/external_dual_three_phase_health_features.csv.gz	cef907bd7c7fef66170a87672df937827bd9a0e8c3f2083599ab277a19de9f0e
sampling input	data/processed/kaist_current_features_10khz.csv.gz	4d196d756b01ab43bd6431a2334a5922787eaf83df7836927c4f1d4d436b13ad
raw-file hash registry	data/processed/external_dual_three_phase_health_features.csv.metadata.json	f5a2925970af69db85caa3699841600b4f9457396ba7250078147a8beb9448ad
primary result	results/external_pmsm_validation/aggregate_summary.csv	d21ae9068e39ecfb0c3d4b597fd150f490079ccd0739bfc4b14a79cdad38cfa
paired diagnostic	results/external_failure_diagnostics/paired_vs_proposed.csv	7a5794bb4470c1bd97a17776b40ed469eac8322270dea79d1e5f84410cb1122b
horizon diagnostic	results/external_failure_diagnostics/horizon_detection_record_any.csv	658e1136d0d3cbac95b7a261139fbf4730e93dc41332e505321bf5fb78db25c4
position diagnostic	results/external_failure_diagnostics/block_auroc.csv	225cc74409c08cedeb292eadd3c664381062aab180c36cd93c402d8c61e969a5
seed diagnostic	results/external_seed_sensitivity/aggregate_seed_ranges.csv	f8634fc426c6bed9091a2cdd246b08bd014c74f07992658b2f47da95c182a8bb
feature diagnostic	results/external_feature_drift/feature_diagnostic_summary.csv	6eda1b2d27f262d23782774d58c9887ae1161e58e84499c94eca9e1349fc1720
geometry check	results/external_feature_drift/score_reconstruction_checks.csv	1607861e42f0dbd0ad10422c64dc42a0551e7bf159a2e7af2d0ab138b10eac18
sampling diagnostic	results/sampling_rate_sensitivity/external_method_comparison.csv	77697abc45ab32648118e23f468b499e5b52c83ac73b94aa4ccb8ed2931373

The external processed-feature metadata JSON contains the individual SHA-256 hash, sampling rate, length, health/fault metadata, and subsystem/window counts for all 56 MAT records. The hash registry above fixes the inputs and derived result tables from which this supplement was mechanically generated.

S8.1. Limits on inference

- **One external motor.** The eight loads, six fault-turn configurations, two subsystems, and eight ordered blocks are repeated conditions on one physical dual-three-phase motor. Neither record bootstrap nor Holm correction creates independent motor replication.
- **Dependent calibration and test blocks.** The 24 calibration blocks come from three healthy load records and the 32 held-out health blocks from four records. Conformal p-values and Wilson intervals are therefore empirical/descriptive, not guarantees under arbitrary within-record dependence.
- **Post-reveal diagnostics.** Sections S3-S7 were specified after external fault reveal. They retain the frozen predictions or mechanically rerun an explicitly defined sampling arm; they introduce no new selected model, feature, ridge, threshold, aggregation, or primary claim.
- **Speed-position coupling.** The analyzed record segment is an acceleration ramp. Block ID, time, electrical-frequency proxy, and speed co-move, so first-alarm and feature associations cannot isolate causal effects or stationary detection delay.
- **Current-only scope.** The pipeline reads only the two three-phase current sets. Voltage, dq variables, and other modalities were not used. The test therefore does not cover faults whose detectable signature requires those channels.
- **Dataset-specific target adaptation.** Target-only methods use 60 healthy adaptation windows per subsystem, while source-balanced variants combine these with source healthy references. Performance gaps describe this locked protocol, not every possible transfer strategy.

The decisive next validation is a prospectively frozen replication on additional independent external motors, preferably with stationary speed/load segments as well as ramps. Until that experiment, external uncertainty is conditional on the single observed motor and the failure of source balancing should not be generalized as a population-level effect.

S9. Prospectively logged secondary transient-set audit

This audit used Zenodo 10.5281/zenodo.15631383: 12 records from one 200 W PMSM and nine records from one 20 kW PMSM, all sampled at 10 kHz and all containing a transition from prefault operation to an interturn short circuit. It has no separate healthy-only records. Signal values remained sealed until the parser, onset rule, whole-record split, 80% motor-compatibility gate, detector family, and outputs had been frozen.

S9.1. Compatibility gate and preserved primary failure

Stage	Motor	Compatible records	Fraction (%)	Use
Frozen primary parser	200W	0/12	0.00	no features or scores
Frozen primary parser	20kW	0/9	0.00	no features or scores
Post-reveal implicit-time repair	200W	12/12	100.00	descriptive sensitivity only
Post-reveal implicit-time repair	20kW	4/9	44.44	44.44%; failed the frozen 80% gate

The frozen primary parser required an explicit monotonic 10 kHz time candidate. All 21 MATLAB `timeseries` objects instead stored uniform timing in `TimeInfo` while their explicit `Time_` arrays were empty. The run therefore stopped before feature extraction and produced no detector scores. After that failure was committed, a separate opt-in parser reconstructed time from the stored start, increment, and length. Five 20 kW records then remained incompatible because detected onset overlapped the frozen 0.5 s baseline. No alternative baseline or record subset was selected, so the 20 kW motor has no quantitative endpoint.

S9.2. Post-reveal 200 W single-motor sensitivity

Each of 12 200 W records was held out in turn. Depending on the hash split, 6 other records supplied prefault fit windows and 5 records supplied 67-88 calibration windows; the held-out record supplied 176 prefault test windows, five primary first-second windows, and ten windows over the full 2 s post-onset horizon.

Method	Held-out prefault alarms	First 1 s fault alarms	Record-any alarms	Full 2 s fault alarms	Mean record AUROC
Target Ledoit-Wolf	7/176	0/60	0/12	0/120	0.280
Target sample covariance	11/176	0/60	0/12	0/120	0.562
Target Log-Euclidean covariance	11/176	0/60	0/12	0/120	0.546
Source covariance	9/176	0/60	0/12	0/120	0.532
Entity-balanced covariance	13/176	0/60	0/12	0/120	0.559
Proposed: Log-Euclidean entity covariance	8/176	0/60	0/12	0/120	0.541
Target OC-SVM (RBF)	9/176	0/60	0/12	0/120	0.389
Source+target OC-SVM (RBF)	8/176	0/60	0/12	0/120	0.161
Target Isolation Forest	11/176	0/60	0/12	0/120	0.447
Source+target Isolation Forest	12/176	0/60	0/12	0/120	0.524
Target MinCovDet	9/176	0/60	0/12	0/120	0.564
Source+target MinCovDet	9/176	0/60	0/12	0/120	0.539

Every method produced zero thresholded alarms in both post-onset horizons. Held-out prefault alarms ranged from 7/176 to 13/176. Proposed produced 8/176 prefault alarms and mean record AUROC 0.541; Target MinCovDet produced 9/176 and AUROC 0.564. Because all 12 methods tie at zero detection, paired thresholded transfer comparisons are uninformative; score-ranking differences do not rescue the missed early alarms. This analysis is post-reveal, contains one physical motor, and reuses prefault and fault segments from the same transition records. It cannot confirm cross-capacity transfer or support population inference.

S9.3. Audit hashes

Role	Artifact	SHA-256
prospectively frozen pre-reveal protocol	docs/secondary_transient_validation_protocol.md	d5f072dd6b494e6b09c1ec6c8956c5d0e60310fc724711d4a66792a78b1b8a08
chronological reveal log	docs/secondary_transient_reveal_log.md	2620b53b6a6ef16d093d74f9b7522fd937dff8c3d61f6bef2ecbb515ad374801
frozen parser compatibility	results/transient_feature_build/record_compatibility.csv	b2f737ca6d2dfbd20f211b1e083dd7a045954658f21f7a7553bcf8fb93aa219e
post-reveal compatibility	results/transient_feature_build_post_reveal_implicit_time/record_compatibility.csv	315d0887fc8db78d87f6ac8db3c120bc91977b4ee4b4e42a8b65748a5c7172a0
post-reveal features	data/processed/transient_pmsm_features_post_reveal_implicit_time.csv.gz	92fea2691ea737319dd43469130b0275bce596bab1e98f0bc4c391f109899ad1
200 W method summary	results/transient_pmsm_validation_post_reveal_200w/aggregation_summary.csv	aa4904f52069c94e3d5ddf885b63fffa54cd35b71671fc5ab85cc9ee1bede41b
200 W record summary	results/transient_pmsm_validation_post_reveal_200w/per_record_summary.csv	b73c1260342d09d1d52546d6fbded7106e352127e80e6dacc67538d556447ad8
200 W predictions	results/transient_pmsm_validation_post_reveal_200w/window_predictions.csv.gz	4c474ac9da88f2b34daf2bb2151939ccc2b77b004ea3e4dfb959fe5da19e3d6

The immutable frozen parser failure and the opt-in repaired analysis are stored in separate result directories. The repaired feature-table hash is checked against both the extractor and validation metadata before this section is generated.